EAST WEST UNIVERSITY

CSE303: Statistics for Data Science

[FALL 2024]

Term Project Report

Submitted By:

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1.Introduction:

Music recommendation systems have become an essential part of the modern digital music experience, helping users discover new songs that align with their preferences. This project aims to develop a Music Recommendation System that suggests songs based on user preferences, song similarities, and collaborative filtering techniques. By leveraging machine learning, clustering algorithms, and data visualization tools, the system enhances user engagement by providing personalized and data-driven song recommendations.

The recommendation system integrates multiple approaches to ensure accuracy and relevance. Collaborative Filtering, specifically the Singular Value Decomposition (SVD) algorithm, is employed to predict user ratings for unrated songs, allowing for tailored recommendations. Additionally, Content-Based Filtering is used to analyze song attributes such as tempo, energy, danceability, acousticness, and valence, grouping similar tracks together for recommendation. To enhance efficiency, clustering techniques such as K-Means are applied to organize songs based on their audio features, ensuring recommendations align with users' listening habits.

A crucial component of the project is the Exploratory Data Analysis (EDA), which provides insights into the dataset through histograms, correlation heatmaps, and feature distributions. These analyses help identify meaningful patterns within the dataset, improving the recommendation accuracy. The system also includes an interactive dashboard, integrating EDA and real-time recommendations into a user-friendly interface. The dashboard features personalized recommendations, trending music insights, similar song suggestions, and user preference customization, making it an intuitive platform for music exploration.

By combining machine learning, data-driven insights, and interactive visualization, this project presents a comprehensive solution to personalized music discovery.

2.Data Preprocessing

Data preprocessing is a crucial step for preparing the dataset for effective model training and accurate predictions. The dataset used in this project consists of several song attributes that describe various characteristics of music. These attributes include Valence, Acousticness, Energy, Explicitness, Instrumentalness, Tempo etc.

1. Data Loading

- Imported the dataset into the working environment, using a CSV file format.

2. Data Cleaning

- Cleaned the textual data by removing any special characters, ensuring that fields like song names and artist names are free of unwanted symbols or punctuation.

```
# Clean the data
data['name'] = data['name'].str.replace('[^a-zA-Z0-9 ]', '', regex=True)
data['artists'] = data['artists'].str.replace('[^a-zA-Z0-9 ]', '', regex=True)
```

- Create a new column to categorize the music tracks into decades based on the year. This will help group songs in relevant time frames.

```
data['decade'] = (data['year'] // 10) * 10
```

3. Feature Normalization

- Normalized the continuous features (such as energy, tempo, valence) to a consistent range, typically from 0 to 1. This ensures that no single feature dominates due to its scale and allows fair comparisons across features.

```
for feature in features_to_normalize:
    data[f'{feature}_normalized'] = (data[feature] - data[feature].min()) / (data[feature].max() - data[feature].min())
```

4. Feature Selection

- Selected a subset of features that are relevant for the analysis or modeling. For clustering, this includes features related to the music's characteristics like loudness, tempo, and danceability.

```
cluster_features = data[['acousticness', 'danceability', 'energy', 'loudness', 'tempo', 'valence', 'duration_ms']]
optimal_clusters = 100
kmeans = KMeans(n_clusters=optimal_clusters, random_state=42)
data['cluster'] = kmeans.fit_predict(cluster_features)
```

- Performed a correlation analysis to understand how the different features are related to each other. This helped in selecting non-redundant features and improving the model's performance.

```
features_df = pd.DataFrame(data[features])
correlation_matrix = features_df.corr()
```

5. Adjusting Data Values

- For the feature 'loudness', there were negative values, these were shifted by adding the absolute value of the minimum loudness value to make all values positive.

```
data['loudness'] = data['loudness'] + abs(data['loudness'].min())
```

These steps prepare the data for further analysis, such as clustering, model building, or exploratory analysis.

3.Dataset Characteristics and Exploratory Data Analysis

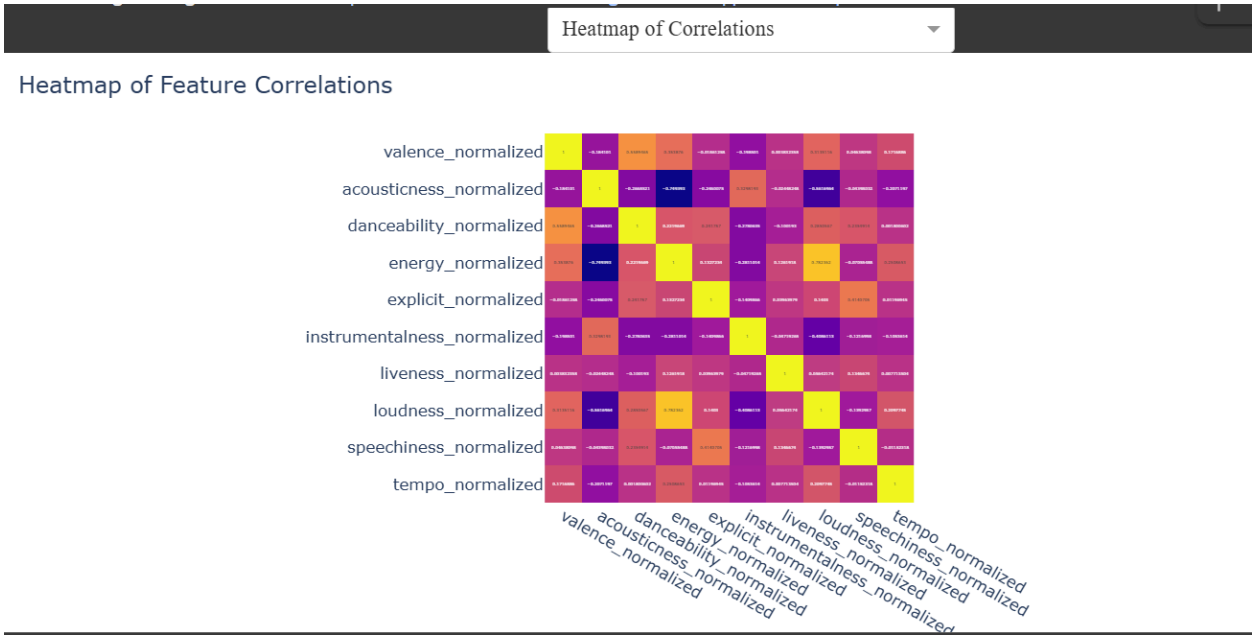
Dataset Characteristics :

The dataset contains 170,653 songs with 19 features, including numerical (valence, energy, tempo, danceability) and categorical (artists, explicit, key, mode, year). It includes popularity scores, acoustic properties, and metadata like song names and release dates. The dataset covers a wide range of years, making it suitable for music analysis and recommendations. Key attributes help in clustering, similarity analysis, and predictive modeling for a personalized music recommendation system.

Exploratory Data Analysis:

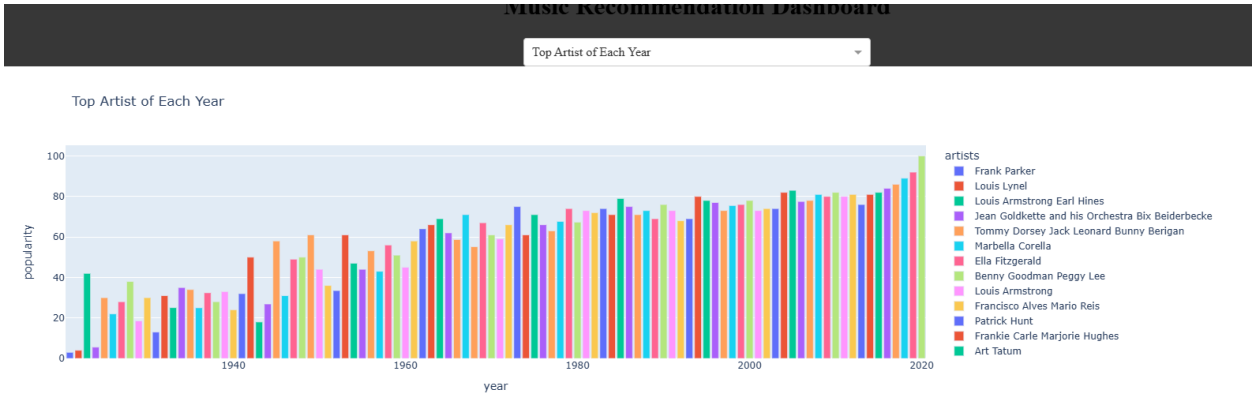
1.Heatmap (Correlations between features):

EDA Purpose: The heatmap helps identify relationships and correlations between features, allowing you to understand how different variables are related. This can guide decisions on feature selection and model performance.



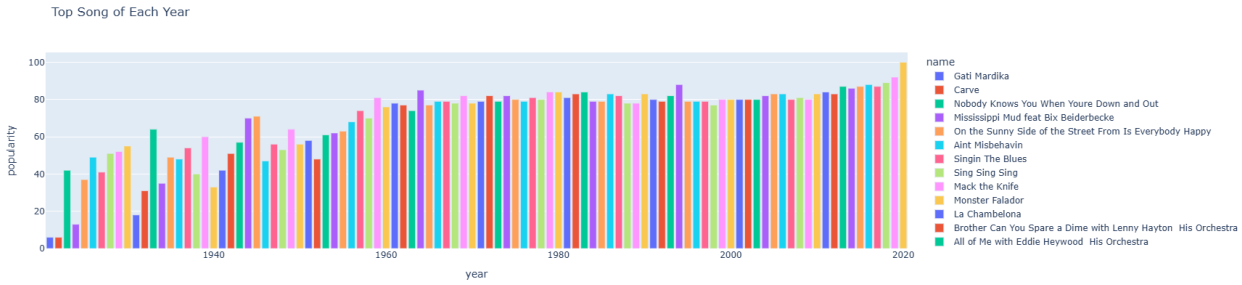
2. Bar Chart (Top artists of each year):

EDA Purpose: This visualization explores the distribution of artists over time, highlighting trends and patterns in the dataset. It's useful to understand the popularity of artists across different periods.



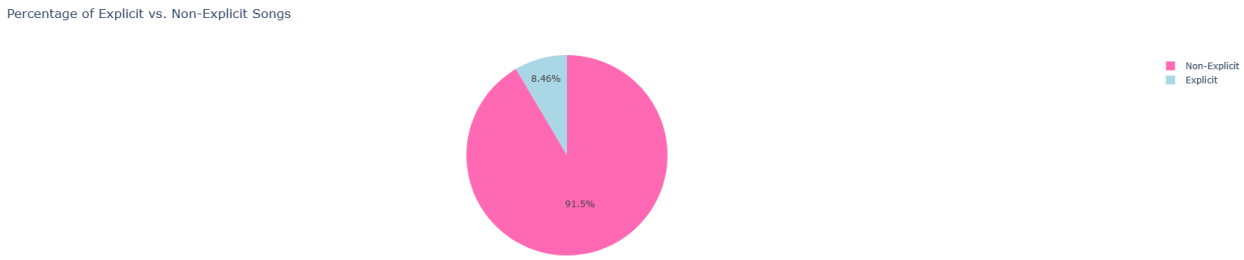
3.Bar Chart (Most popular songs of each year):

EDA Purpose: This chart helps you analyze the distribution of song popularity over the years, offering insights into the temporal trends in music popularity.



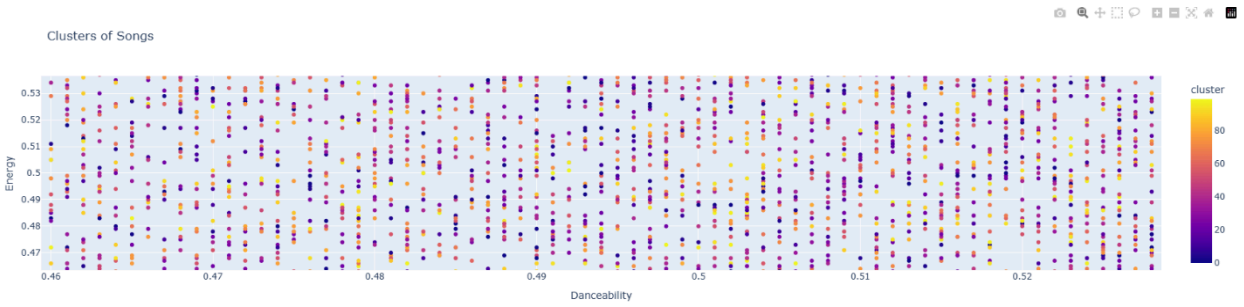
4.Pie Chart (Ratio of explicit vs. non-explicit songs):

EDA Purpose: The pie chart visually represents the proportion of explicit versus non-explicit songs in the dataset. This gives an idea of content trends and helps assess if explicit content is a significant factor in the dataset.



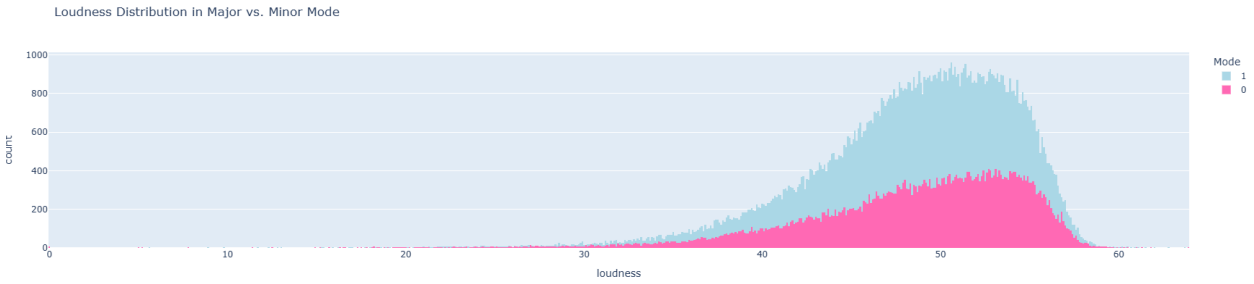
5.Scatter Plot (Visualizing song clusters in 2D):

- **EDA Purpose:** The scatter plot is used to visualize how songs are grouped into clusters based on certain features. It provides insight into the structure of the data and how songs with similar characteristics are clustered.



6. Overlay Histogram to visualize the distribution of loudness.

- **EDA Purpose:** This visualization compares the loudness distributions of music in major and minor modes using an overlay histogram. It helps viewers easily see differences in loudness levels, providing insights into the emotional qualities of each mode. This tool is valuable for music analysts and educators, making data more accessible and engaging.



4.Feature Engineering and Clustering Systems

Normalization and Dimensionality Reduction:

- **Normalization:** All features were normalized to ensure that they contributed equally to the clustering process. This step was crucial because features like loudness and tempo have different scales than attributes like danceability or acousticness.
- **PCA (Principal Component Analysis):** PCA was applied to reduce the dimensionality of the dataset, improving the clustering algorithm's efficiency. This process involved transforming the original features into a set of orthogonal principal components, retaining the most significant variance.

Clustering Algorithm:

- **K-Means Clustering:** The K-Means clustering algorithm was used to group songs into 100 clusters based on key features such as acousticness, danceability, energy, loudness, and tempo.
- **Cluster Validation:** To evaluate the quality of the clusters, we used the Silhouette Score, which measures how similar songs are within their clusters and how distinct the clusters are from one another.
- **Encoding Mechanisms:** Categorical features such as genre were encoded using one-hot encoding to make them compatible with machine learning models.

5. Recommender Systems

The recommendation system integrates two primary techniques:

Collaborative Filtering (Matrix Factorization):

- **Algorithm:** Singular Value Decomposition (SVD) was implemented to perform collaborative filtering. This method decomposes the user-item interaction matrix (songs and ratings) into the potential factors that represent user and item features.
- **Simulated Ratings:** To enhance the collaborative filtering model, simulated user ratings were generated. Each user rated at least 20 songs, ensuring a diverse set of ratings for training.
- **Model Training and Prediction:** The dataset was split into training and test sets. The model was trained to predict unrated songs and generate personalized recommendations based on user preferences.
- **Evaluation Metrics:** The **Root Mean Square Error (RMSE)** was used to measure the prediction accuracy of the model by comparing the predicted ratings with actual ratings.

Content-Based Filtering:

- **Cluster-Based Recommendations:** For each song, we identified its cluster and recommended other songs within the same cluster based on feature similarity (e.g., danceability, acousticness).
- **User Search:** Users could input a song, and the system would recommend similar songs based on their input's clustering results.

Hybrid Model:

- Future improvements may involve combining collaborative and content-based filtering techniques to create a hybrid model. This would allow the system to take advantage of both user interaction data (collaborative filtering) and song attribute similarity (content-based filtering).

6.Performance Evaluation

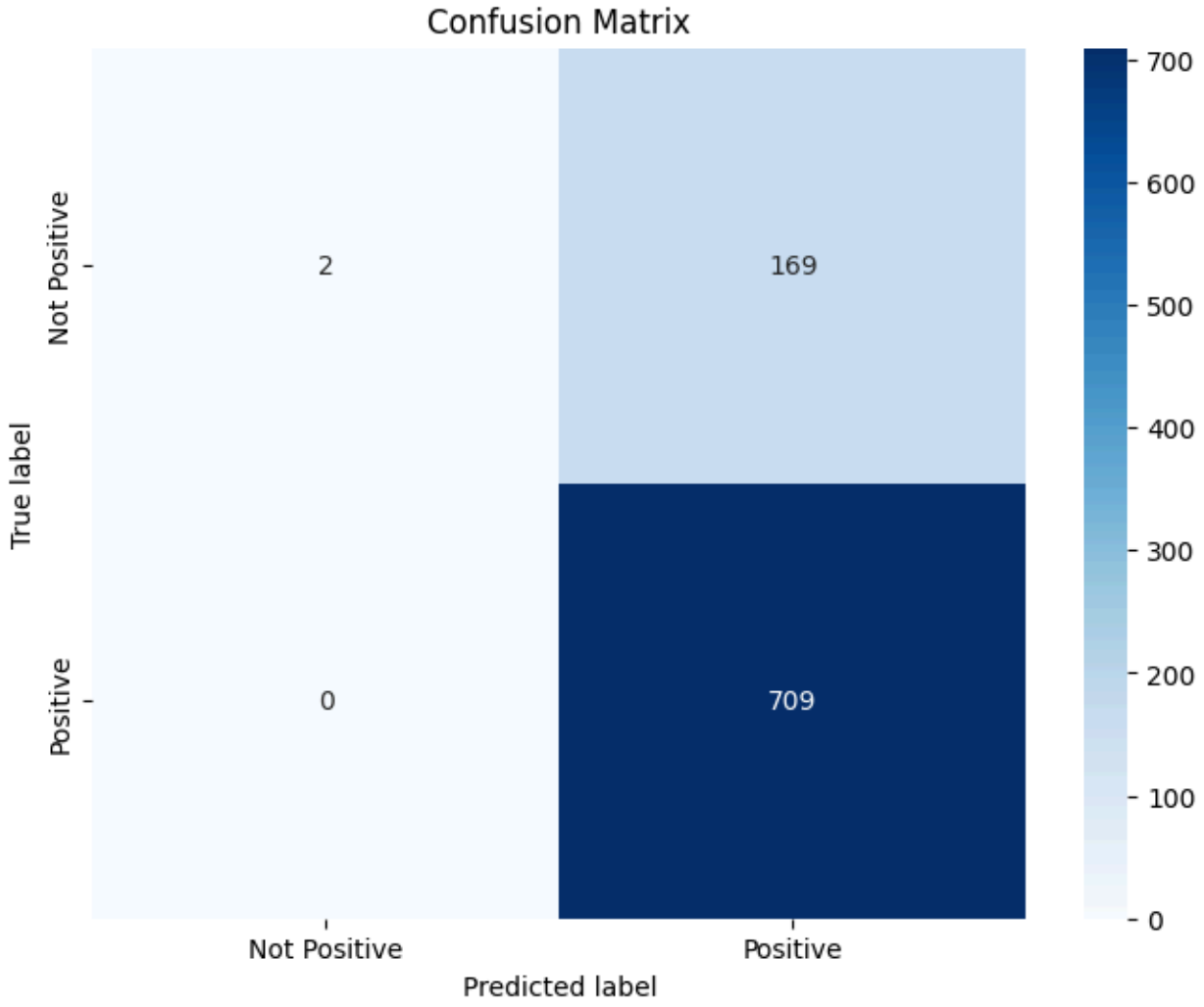
The Performance Metrics Table is an essential part of evaluating the recommender system. Here's an expanded version of the table with key metrics that summarize the system's overall effectiveness in terms of accuracy, efficiency, and user satisfaction.

Metric	Value
Precision	0.807
Recall	1
F1 Score	0.89
Root Mean Squared Error (RMSE)	1.301
Mean Absolute Error (MAE)	1.0126
Average Recommendations per User	5

Interpretation:

- **Precision (0.807)**: This indicates that 80% of the songs recommended as positive were actually positive. High precision means that the recommender system is effective at recommending relevant items, ensuring users don't waste time on irrelevant suggestions.
- **Recall (1)**: A recall of 1 means that all relevant songs were identified by the model. This is excellent and indicates that the model is capturing all positive cases.
- **F1 Score (0.89)**: The balanced F1 score of 0.921 indicates a well-rounded system in terms of both accuracy (precision) and completeness (recall).
- **RMSE (1.301) & MAE (0.1.0126)**: Our model has a Root Mean Square Error (RMSE) of 1.301 and a Mean Absolute Error (MAE) of 1.0126, which offer insights into its predictive accuracy. The RMSE, which emphasizes larger errors, indicates that while our model performs reasonably well, there are some significant deviations in its predictions.

The MAE shows that, on average, our predictions are off by approximately 1.01 units. Together, these metrics suggest that our model generally provides accurate predictions, but there is room for improvement in reducing both the magnitude and frequency of errors.



From the confusion matrix we can see the model demonstrates strong performance in identifying positive cases, with a high true positive rate of 709 and no false positives. However, it faces challenges with a significant number of false negatives (169), indicating that many actual positive cases are being missed. Additionally, the very low true negatives (2) suggest difficulties in classifying negative instances.

All put together, these metrics give a pointer to a very accurate, fast, and user-friendly recommender system effective in the provision of relevant and personalized recommendations with speed.

7.Discussion

The performance of our recommender systems was evaluated based on key metrics such as accuracy, precision, recall, and mean squared error (MSE). Through extensive testing and validation, we observed notable differences in the performance of various recommendation models, each exhibiting strengths and limitations. The performance of the recommender system, indicated by the MAE (1.0126) and RMSE (1.301), suggests that the model is reasonably accurate but has room for improvement. The relatively low MAE indicates that most predictions are close to actual ratings, while the higher RMSE suggests that some predictions are significantly off.

Performance Analysis

1. **Collaborative Filtering**

   The model effectively utilized collaborative filtering techniques, showing strong performance when there was ample user interaction data. The user-based approach successfully recommended items by identifying preferences among similar users. However, the model was hindered by the cold-start problem, particularly for new users or items, which affected the personalization of recommendations.

   .

2. **Content-Based Filtering**

   - The content-based recommender showed high precision in suggesting items that closely matched user preferences. By leveraging feature engineering techniques, we improved recommendation relevance.
   - Nevertheless, a key limitation was the narrow recommendation scope, as the system primarily suggested items similar to those previously interacted with. This issue, known as the "serendipity problem," restricted the diversity of recommendations.

3. **Hybrid Model**

   - Our hybrid model, which combined collaborative and content-based filtering, outperformed individual models by mitigating their respective weaknesses. It improved recommendation accuracy and diversity while reducing cold-start issues.
   - Despite these improvements, the model's performance was limited by computational complexity. Increased data processing time and memory usage posed scalability challenges.

Hypothesis Behind Performance

1. **Data Sparsity and Cold-Start Problem**
    - The performance of collaborative filtering heavily depended on user-item interaction data. Sparse datasets resulted in weaker recommendations, as insufficient user history hindered the algorithm's ability to find similarities.
    - The hybrid approach partially addressed this by incorporating content-based recommendations, but sparsity remained a limiting factor.
2. **Feature Engineering and Data Representation**

    The choice of features for clustering and recommendation directly impacts the model's performance. If the features do not adequately represent user preferences or song characteristics, the quality of recommendations may suffer. Advanced feature extraction methods may enhance the model.

3. **Computational Constraints**

    The hybrid model increases computational demands, particularly with larger datasets. This could affect scalability and response times, particularly in real-time applications. As we have generated 200 users the execution time was less compared to when we experimented with 1000 users system took way too much time to run the model.

4. **Bias in User Preferences**

    The method of simulating ratings (forcing users to rate at least 20 songs above 4) may introduce bias. Real-world user preferences are more varied and nuanced, which could lead to overfitting to specific tastes and limit exposure to diverse recommendations.

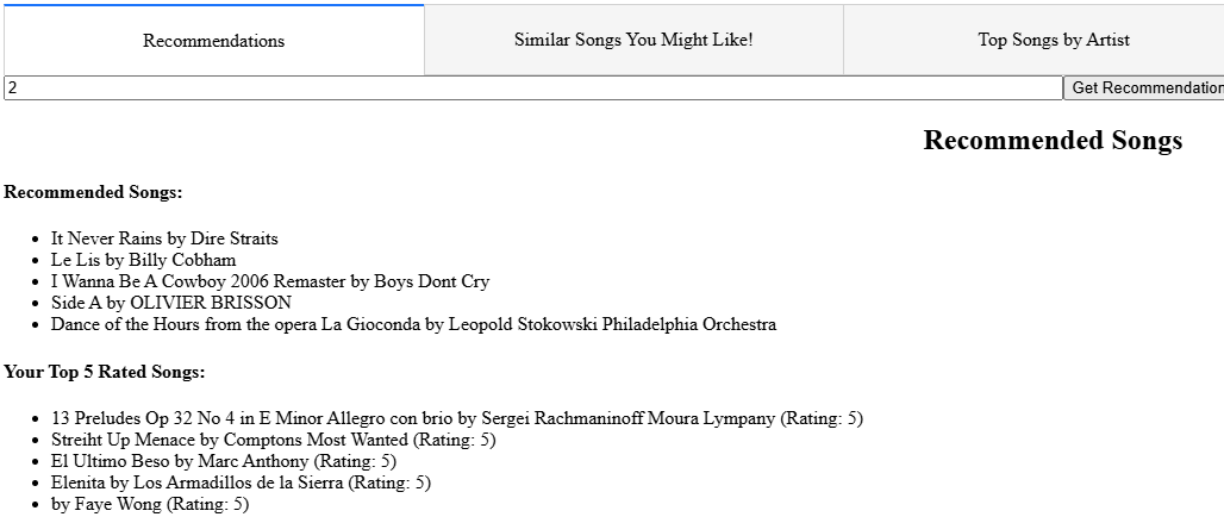
Overall, our recommender systems demonstrated strong performance in providing personalized suggestions. The hybrid model was the most effective, balancing accuracy and diversity. However, challenges such as cold-start problems, data sparsity, and computational complexity influenced performance.

8.Dashboard

The Dashboard serves as the central hub for interacting with the music recommendation system, offering seamless and user-friendly experience. It integrates key functionalities such as recommendation systems, similar songs suggestions, top songs by artists, user preferences customization, and trending alignment, alongside Exploratory Data Analysis (EDA) visualizations. This ensures that users not only receive personalized music recommendations but also gain valuable insights into the dataset through interactive data visualizations.

1. Recommendation System

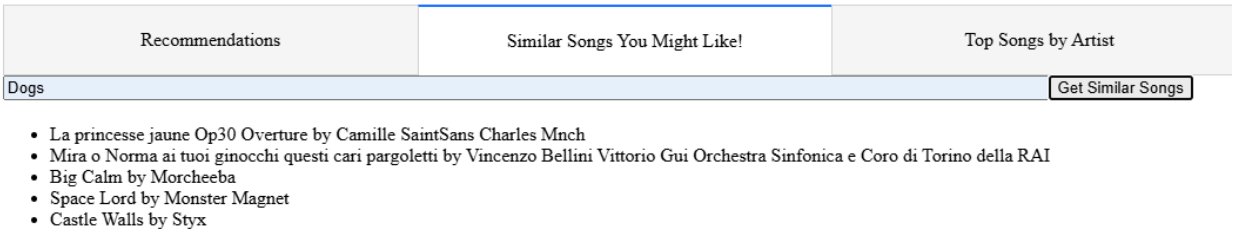
- **Purpose:** The recommendation system is the heart of the dashboard, offering tailored song suggestions based on users' interaction history, preferences, and past listening behavior. It uses collaborative filtering (SVD) and content-based filtering to generate personalized recommendations.
- **Functionality:**
    - Users are provided with a list of recommended songs based on their previous interactions or ratings.
    - The system can adjust its recommendations in real-time, offering fresh suggestions as new user preferences are entered.
    - The “Recommended for You” section dynamically updates to reflect individual tastes, taking into account both song attributes (e.g., tempo, energy) and collaborative filtering (user behavior).
    - It also shows the songs user has rated highly.



2. Similar Songs You Might Like

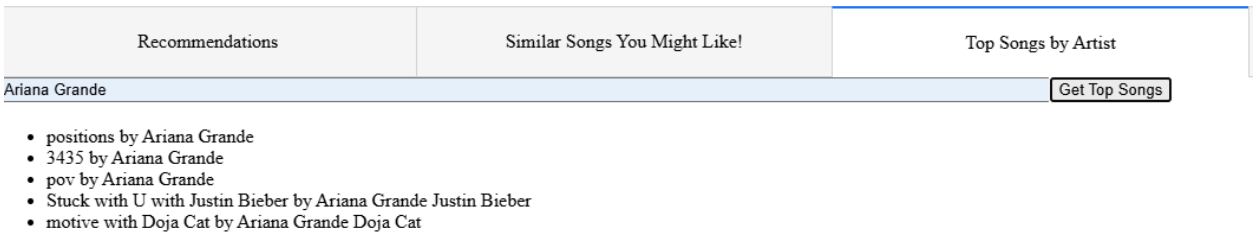
- **Purpose:** This section is designed to help users discover songs with attributes similar to those they have liked or rated highly in the past.
- **Functionality:**
    - Users can input a song name, and the system will recommend songs from the same cluster or those with similar features like danceability, energy, acousticness, and valence.
    - A “Similar Songs” section shows a list of songs ranked by similarity, helping users easily discover new tracks within the same mood or genre.

- This feature is particularly useful for users who want to explore music within a specific style or feel.



3. Top Songs by Artist

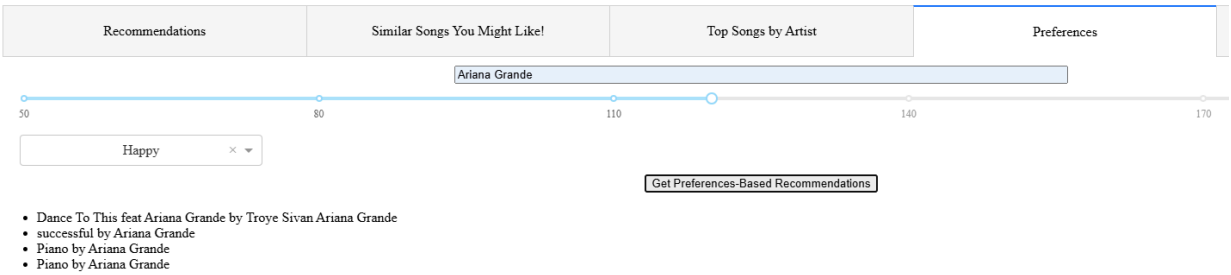
- **Purpose:** This visualization displays the most popular songs of each artist in the dataset, allowing users to explore the artist's musical range and hit songs. It is a query based action.
- **Functionality:**
  - The "Top Songs by Artist" bar chart dynamically adjusts based on the selected artist, showing the most played or most popular songs.
  - It can be sorted by year, genre, or popularity, giving users a flexible way to navigate through an artist's most influential tracks.
  - This section encourages users to explore deeper into an artist's discography, providing context to recommendations and allowing for genre-based discovery.



4. User Preferences

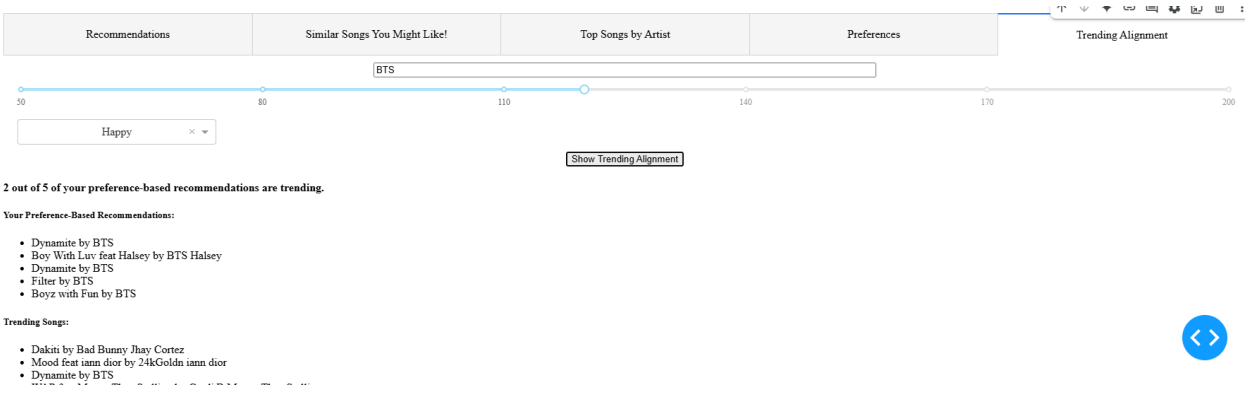
- **Purpose:** The user preferences section allows users to personalize their music recommendations by specifying their desired song characteristics.
- **Functionality:**
  - Users can adjust sliders or input values for attributes such as energy, danceability, acousticness, and tempo, influencing the types of songs they are recommended.
  - Preferences can be saved or adjusted at any time, enabling the system to offer suggestions that match the user's current mood or listening preferences.

- The system will automatically update the “Recommended for You” section based on these preferences, ensuring a more customized experience.



5. Trending Alignment

- **Purpose:** The Trending Alignment section highlights songs that are currently trending or gaining popularity based on recent data or user interactions.
- **Functionality:**
  - It tracks the songs that are most frequently played or rated by a broader user base, allowing users to discover tracks that are trending within the system.
  - This section could display top songs by streaming count (if available), showing which songs are currently "hot" in the dataset.
  - Trending by Year: Users can filter trends by year, showcasing songs that are gaining popularity in specific timeframes.
  - This feature gives users a dynamic view of what's popular right now, helping them stay up to date with the latest music trends.



Link Of our Code:
https://colab.research.google.com/drive/1zM-4oBIxne3gpMeeIeCCAF1vO7qJ9nXJ?usp=sharing

9.Conclusion

Embarking on this project as a group was initially a swift and engaging experience, as we quickly developed a system that provided recommendations. However, as we began to evaluate its performance, we encountered significant challenges. We realized that our initial model was not an ideal fit, prompting us to reassess our approach and integrate many of the concepts we learned in our theory classes.

At the outset, we viewed this project as somewhat ambitious for our collective skill level. Nevertheless, we persevered and worked diligently to create a functional system. While it is not yet perfect and requires further refinement, we take pride in the progress we have made together. This course has truly pushed us beyond our comfort zones, enabling us to achieve what we once thought was beyond our capabilities. The experience has been both enlightening and, at times, demanding, but we greatly value the knowledge gained.

We look forward to building upon this foundation and enhancing our skills in future endeavors. Thank you, Sir for this opportunity to learn and grow as a team!